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- ICES. 2025b. Update of reference points of turbot (*Scophthalmus maximus*) in Subarea 27.4 (North Sea) for the ICES Benchmark Workshop on selected North Sea and Celtic Sea stocks (WKBNSCS 2025). ICES Working Document. 5 pp.
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20.14 Tables and figures

Table 20.1. Turbot in Subarea 4. ICES estimated landings (tonnes) SOP corrected and discards of turbot.

Year	Landings	Landing SOP	Discards
1981	4755	1	
1982	4453	1	
1983	4575	1	
1984	5297	1	
1985	6188	1	
1986	5263	1	
1987	4271	1	
1988	4041	1	
1989	4927	1	
1990	5750	1	
1991	6340	-0.007	
1992	5933	-0.007	
1993	5546	-0.008	
1994	5244	-0.008	
1995	4671	-0.009	
1996	3644	-0.011	

Year	Landings	Landing SOP	Discards
1997	3382	-0.012	
1998	3086	1	
1999	3187	-0.012	
2000	4025	-0.009	
2001	4100	-0.009	
2002	3749	-0.010	
2003	3374	1	
2004	3317	1	
2005	3195	1	
2006	2976	1	
2007	3509	1	
2008	3005	1	
2009	3089	1	
2010	2692	1	
2011	2771	1	
2012	2914	1	
2013	2982	1	97
2014	2834	1	158
2015	2922	1	112
2016	3493	1	666
2017	3441	1	496
2018	3140	1	486
2019	3046	1	230
2020	3104	1	199
2021	2659	1	141
2022	1691	1	49
2023	1662	1	195
2024	1750	1	81

Table 20.2. Turbot in Subarea 4. Official estimated landings (tonnes) for each country participating in the fishery.

Year	Netherlands	UK	Denmark	Belgium	France	Germany	Norway	Other	Total
1975	3349	503	387	158	21	169	0	1	4588
1976	3253	631	588	146	38	156	0	2	4814
1977	2973	683	474	145	37	172	0	0	4484
1978	3196	752	693	170	50	173	0	< 1	5034
1979	3999	838	1164	187	22	151	0	3	6364
1980	3241	559	1360	162	17	146	0	0	5485
1981	3073	404	1044	142	6	86	0	0	4755
1982	3029	335	880	153	14	42	0	< 1	4453
1983	3163	277	893	174	24	44	0	< 1	4575
1984	3800*	282	886	242	40	46	0	1	5297
1985	4600*	312	983	222	37	34	0	0	6188
1986	3810*	287	997	133	5	31	0	0	5263
1987	2760*	345	988	130	21	27	0	0	4271
1988	2660	328	858	129	24	41	0	1	4041
1989	3666	333	637	176	30	85	0	0	4927
1990	3732	437	1046	292	52	184	0	7	5750
1991	3780	688	1233	350	64	186	30	9	6340
1992	3495	902	907	317	81	163	65	3	5933
1993	2939	1013	817	355	123	252	47	0	5546
1994	2724	882	862	330	141	263	42	0	5244
1995	2476	703	761	315	108	275	33	0	4671
1996	1776	687	618	210	160	157	36	0	3644
1997	1854	619	479	169	1	215	45	0	3382
1998	1695	582	392	198	22	164	33	< 1	3086
1999	1808	488	411	224	0	224	32	0	3187
2000	2280	549	469	302	21	349	55	0	4025
2001	2226	642	506	333	17	297	79	0	4100

Year	Netherlands	UK	Denmark	Belgium	France	Germany	Norway	Other	Total
2002	1898	551	677	243	15	280	85	0	3749
2003	1893	431	486	192	18	289	65	0	3374
2004	1762	463	518	207	15	278	74	0	3317
2005	1903	347	429	159	18	274	65	0	3195
2006	1828	381	338	147	22	221	40	< 1	2976
2007	2263	485	310	173	32	203	43	0	3508
2008	1744	370	457	182	21	199	32	< 1	3005
2009	1698	421	548	172	24	198	29	< 1	3090
2010	1469	386	467	118	37	191	26	< 1	2695
2011	1540	396	548	122	33	144	28	< 1	2811
2012	1740	366	481	150	30	187	37	< 1	2991
2013	1763	374	498	161	40	219	29	< 1	3085
2014	1594	391	433	176	42	197	38	< 1	2872
2015	1739	336	392	215	46	236	10	4	2978
2016	1854	404	505	339	38	273	8	1	3422
2017	2118	397	486	336	40	252	13	1	3641
2018	1914	368	331	267	27	306	15	1	3231
2019	1901	362	273	228	14	326	13	1	3121
2020	2099	354	257	161	5	297	17	1	3192
2021	1683	325	326	134	3	222	13	1	2705
2022	926	210	333	90	1	125	16	1	1702
2023*	828	238	255	125	< 1	172	28	1	1647
2024*	835	281	230	158	< 1	207	41	1	1754

* Preliminary

Table 20.3. Turbot in Subarea 4. Observed landings in numbers (units: thousands) SOP corrected.

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.000	297.536	751.303	529.395	455.757	174.143	66.672	106.476

Year	Age							
	1	2	3	4	5	6	7	8+
1982	0.000	168.924	1045.530	266.880	166.925	291.869	97.956	154.930
1983	0.000	392.746	657.507	467.973	107.468	110.399	175.856	198.327
1984	0.000	1244.974	1174.848	298.755	150.819	57.638	54.756	187.323
1985	0.000	732.041	2223.746	602.207	164.824	100.368	23.941	147.329
1986	0.000	353.580	1400.976	664.272	174.407	63.854	27.638	118.178
1987	12.262	611.207	514.999	637.618	149.029	49.048	17.921	66.025
1988	31.063	935.353	773.996	153.591	151.865	77.658	24.160	66.441
1989	0.000	641.708	1120.288	340.300	150.164	78.863	30.249	65.899
1990	42.868	954.063	1028.833	304.064	159.509	72.776	97.699	109.662
1991–1997	NO DATA							
1998	0.000	459.781	985.976	405.289	82.590	33.462	9.621	16.186
1999–2002	NO DATA							
2003	259.774	2363.264	570.141	367.770	87.564	40.766	25.588	25.394
2004	453.290	2063.262	825.674	144.077	85.892	10.067	7.850	6.326
2005	380.154	2191.337	797.918	254.655	27.424	24.160	2.873	21.221
2006	884.902	1645.164	807.535	119.124	35.110	7.900	16.176	18.132
2007	76.706	2715.900	601.933	278.406	39.361	28.417	8.064	15.543
2008	176.429	1342.581	816.641	218.982	194.120	46.856	12.814	10.165
2009	110.457	1016.694	949.607	410.079	86.929	24.472	10.772	18.104
2010	281.699	1419.201	390.191	312.867	173.683	89.102	30.930	19.771
2011	210.757	1940.194	602.162	110.621	137.555	76.954	32.224	23.577
2012	0.000	1911.605	777.988	267.077	42.511	63.987	73.107	24.751
2013	162.129	1484.665	1015.946	305.667	85.457	24.408	39.459	24.317
2014	68.896	391.794	650.547	683.844	137.556	121.935	38.029	105.115
2015	40.938	1265.013	483.799	339.712	329.270	114.230	44.945	82.995
2016	0.000	1125.497	1075.877	360.984	387.787	202.800	48.855	76.423
2017	5.507	263.098	1324.693	478.283	110.665	49.955	78.228	48.401
2018	187.291	733.128	494.694	948.980	263.545	71.155	47.308	74.676

Year	Age							
	1	2	3	4	5	6	7	8+
2019	172.159	1061.728	881.441	262.642	358.720	122.170	22.880	63.883
2020	209.328	1549.632	822.228	385.818	141.071	142.926	41.532	40.699
2021	0.000	652.895	929.355	483.655	182.535	62.664	66.885	59.057
2022	181.998	394.750	438.382	222.792	124.784	30.715	11.201	36.945
2023	99.565	553.734	398.675	228.952	140.761	73.281	25.148	27.836
2024	323.257	509.610	572.631	206.068	96.528	59.620	22.190	37.184

Table 20.4. Turbot in Subarea 4. Raw weights-at-age in the landings (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	NA	0.90	1.00	1.70	2.60	3.60	4.40	6.86
1982	NA	0.90	1.10	1.80	2.60	3.20	4.50	5.56
1983	NA	0.90	1.20	2.00	2.80	3.60	4.00	5.52
1984	NA	0.80	1.30	2.20	3.20	3.80	4.50	6.24
1985	NA	0.70	1.10	2.00	3.20	4.20	5.00	6.49
1986	NA	1.00	1.30	2.10	3.00	3.70	6.30	6.43
1987	0.70	1.10	1.60	2.10	3.80	4.60	6.10	7.96
1988	0.70	1.00	1.60	2.80	3.10	4.60	6.00	6.66
1989	NA	1.00	1.50	2.70	3.90	4.70	6.90	7.91
1990	0.90	1.00	1.60	2.70	3.40	5.40	5.60	7.37
1991–1997	NO DATA							
1998	NA	0.83	1.26	2.12	3.34	4.92	5.38	6.83
1999–2001	NO DATA							
2002	0.41	0.65	1.29	1.86	2.71	3.61	3.61	4.01
2003	0.5	0.62	1.15	1.78	2.24	2.74	2.59	4.98
2004	0.43	0.69	1.2	2.12	3.17	3.76	5.15	8.61
2005	0.44	0.62	1.13	1.89	2.89	3.47	4.6	4.96
2006	0.41	0.66	1.31	1.92	3.37	5.09	2.7	2.29

Year	Age							
	1	2	3	4	5	6	7	8+
2007	0.34	0.7	1.25	1.75	3.27	3.72	4.17	4.96
2008	0.37	0.68	1.27	1.78	1.79	2.76	4.91	4.21
2009	0.41	0.62	1.25	1.76	2.95	4.83	5.47	4.01
2010	0.35	0.61	1.07	1.62	2.19	2.67	2.65	4.51
2011	0.48	0.55	1.06	1.79	1.97	3.25	4.48	3.9
2012	NA	0.6	0.91	1.46	2.58	3.01	3.47	3.23
2013	0.61	0.61	1.00	1.64	2.23	3.41	2.27	2.91
2014	0.41	0.59	1.07	1.42	1.67	1.85	3.034	3.09
2015	0.41	0.59	1.1	1.3	1.67	2.12	2.78	3.16
2016	0.00	0.66	0.93	1.33	1.22	1.94	2.93	2.39
2017	0.54	0.98	1.18	1.74	2.15	2.37	3.07	3.51
2018	0.34	0.59	0.98	1.36	1.41	1.9	2.86	2.71
2019	0.44	0.58	0.94	1.5	1.77	2.11	3.63	2.24
2020	0.44	0.63	0.96	1.29	1.48	2.01	2.87	3.13
2021	0	0.57	0.97	1.39	1.57	1.66	2.69	2.4
2022	0.48	0.79	1.13	1.59	1.75	2.46	3.09	3.25
2023	0.37	0.62	1.02	1.5	1.78	1.78	2.8	3.6
2024	0.45	0.7	0.98	1.32	1.92	1.66	2.18	2.68

Table 20.5. Turbot in Subarea 4. Raw weights-at-age in the stock estimated as the catch weights in Q2 (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	NA	0.90	0.80	1.48	2.59	3.23	5.66	6.65
1982	NA	0.59	1.01	1.80	2.53	3.33	4.88	6.19
1983	NA	0.61	1.13	1.99	2.77	3.38	3.97	5.02
1984	NA	0.66	1.04	2.07	2.87	4.25	4.93	6.42
1985	NA	0.59	1.02	1.83	2.95	4.46	5.99	6.13
1986	NA	0.91	1.12	1.98	3.08	3.48	7.02	6.32

Year	Age							
	1	2	3	4	5	6	7	8+
1987	0.7	0.72	1.25	1.87	3.60	3.24	5.36	8.30
1988	0.7	1.16	1.65	2.65	3.31	5.78	7.24	8.05
1989	NA	0.81	1.48	2.96	5.30	5.77	8.26	8.31
1990	0.9	0.84	1.79	3.09	3.02	5.34	3.47	8.78
1991–1997	NO DATA							
1998	NA	0.8	1.03	1.67	3.08	5.06	2.57	7.49
1999–2001	NO DATA							
2002	0.44	0.60	1.28	1.94	2.64	3.68	2.95	4.05
2003	NA	0.50	1.14	1.99	2.45	2.82	4.14	3.43
2004	NA	0.52	1.10	1.90	2.47	2.91	5.35	6.44
2005	NA							
2006	NA							
2007	NA	0.59	1.10	1.57	2.58	2.71	1.72	4.88
2008	NA	0.65	1.14	1.44	2.10	5.16	6.01	7.13
2009	NA	0.44	0.80	1.51	1.65	3.55	4.70	4.97
2010	NA	0.45	1.04	1.62	2.30	2.38	2.71	5.42
2011	NA	0.39	0.95	1.88	2.01	4.00	4.42	5.27
2012	NA	0.51	0.85	1.42	2.20	2.67	2.58	4.18
2013	NA	0.59	0.95	1.60	2.18	3.30	2.51	4.25
2014	0.38	0.57	0.95	1.24	1.50	1.72	1.84	2.97
2015	0.41	0.49	0.89	0.93	1.46	1.40	1.37	4.46
2016	0.41	0.58	0.78	1.30	0.80	1.49	4.78	3.34
2017	0.39	0.38	0.92	1.60	2.04	2.31	2.87	3.24
2018	0.27	0.45	1.03	1.46	1.64	2.72	2.37	4.20
2019	0.44	0.39	0.86	1.37	2.04	2.25	4.25	3.20
2020	0.44	0.56	1.16	1.39	2.39	2.31	3.21	2.80
2021	NA	0.53	0.97	1.49	1.72	1.92	2.76	2.53
2022	0.48	0.74	1.06	1.59	2.01	2.45	2.91	2.61

Year	Age							
	1	2	3	4	5	6	7	8+
2023	0.3	0.63	1.09	1.44	1.76	1.83	2.90	3.42
2024	0.43	0.61	0.92	1.33	1.94	1.57	2.13	2.74

Table 20.6. Turbot in Subarea 4. Modelled weights-at-age in the catch (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.58	0.85	1.10	1.80	2.63	3.55	4.49	6.40
1982	0.58	0.86	1.19	1.86	2.65	3.42	4.48	5.72
1983	0.58	0.86	1.27	2.00	2.80	3.60	4.37	5.63
1984	0.59	0.82	1.30	2.11	3.02	3.82	4.64	5.86
1985	0.63	0.81	1.26	2.10	3.10	4.10	5.04	6.07
1986	0.67	0.95	1.36	2.16	3.14	4.09	5.68	6.22
1987	0.69	1.02	1.55	2.25	3.40	4.43	5.80	7.08
1988	0.70	1.01	1.60	2.56	3.33	4.59	5.94	7.09
1989	0.73	1.01	1.59	2.60	3.62	4.65	6.18	7.47
1990	0.78	1.02	1.61	2.60	3.51	4.88	5.82	7.31
1991	0.72	1.04	1.63	2.54	3.50	4.64	5.81	7.11
1992	0.68	1.02	1.63	2.52	3.45	4.53	5.70	6.98
1993	0.65	0.99	1.61	2.50	3.41	4.45	5.60	6.86
1994	0.62	0.95	1.57	2.46	3.38	4.39	5.51	6.75
1995	0.60	0.91	1.52	2.41	3.34	4.34	5.44	6.65
1996	0.58	0.88	1.46	2.35	3.29	4.30	5.38	6.57
1997	0.56	0.84	1.40	2.26	3.24	4.28	5.32	6.52
1998	0.54	0.82	1.32	2.15	3.15	4.34	5.23	6.50
1999	0.52	0.79	1.30	2.06	2.92	3.93	4.96	6.17
2000	0.49	0.76	1.27	2.00	2.77	3.63	4.59	5.79
2001	0.47	0.73	1.25	1.95	2.68	3.42	4.21	5.34
2002	0.44	0.69	1.24	1.90	2.63	3.36	3.82	4.83

Year	Age							
	1	2	3	4	5	6	7	8+
2003	0.47	0.67	1.19	1.91	2.54	3.19	3.60	4.53
2004	0.44	0.68	1.18	2.00	2.83	3.46	4.31	5.67
2005	0.44	0.66	1.17	1.94	2.90	3.52	4.13	5.29
2006	0.42	0.67	1.20	1.90	2.98	4.01	3.63	4.30
2007	0.39	0.68	1.19	1.80	2.81	3.70	4.24	4.73
2008	0.39	0.66	1.19	1.80	2.34	3.33	4.54	5.23
2009	0.40	0.62	1.15	1.75	2.52	3.56	4.49	5.17
2010	0.39	0.61	1.07	1.68	2.30	3.06	3.62	5.11
2011	0.44	0.58	1.02	1.68	2.17	3.02	3.92	4.76
2012	0.46	0.61	0.96	1.54	2.28	2.85	3.50	4.88
2013	0.50	0.62	0.98	1.51	2.06	2.82	2.91	4.56
2014	0.44	0.63	1.00	1.41	1.81	2.27	3.01	3.70
2015	0.44	0.62	1.02	1.35	1.70	2.20	2.89	3.51
2016	0.46	0.66	0.99	1.40	1.57	2.13	2.88	3.68
2017	0.47	0.74	1.05	1.49	1.81	2.20	2.86	3.53
2018	0.40	0.64	1.01	1.42	1.66	2.12	2.80	3.23
2019	0.43	0.60	0.96	1.42	1.71	2.12	2.96	2.91
2020	0.44	0.62	0.96	1.35	1.62	2.06	2.73	3.07
2021	0.45	0.62	0.98	1.39	1.66	1.96	2.66	2.82
2022	0.45	0.68	1.03	1.47	1.72	2.15	2.72	2.98
2023	0.41	0.64	1.01	1.47	1.77	1.96	2.61	3.00
2024	0.43	0.65	0.99	1.41	1.84	1.94	2.39	2.85

Table 20.7. Turbot in Subarea 4. Modelled weights-at-age in the stock (units: kg).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.52	0.73	0.97	1.67	2.62	3.48	5.07	6.42
1982	0.52	0.65	1.09	1.78	2.60	3.51	4.64	6.18

Year	Age							
	1	2	3	4	5	6	7	8+
1983	0.54	0.64	1.14	1.91	2.75	3.65	4.43	5.76
1984	0.56	0.66	1.12	1.98	2.87	4.03	4.81	6.08
1985	0.61	0.68	1.14	1.96	2.94	4.20	5.37	6.28
1986	0.64	0.79	1.22	2.04	3.06	4.00	5.82	6.73
1987	0.69	0.82	1.36	2.13	3.34	4.12	5.63	7.57
1988	0.70	0.93	1.53	2.47	3.46	4.89	6.11	7.78
1989	0.72	0.88	1.57	2.65	3.90	4.97	6.27	8.01
1990	0.75	0.87	1.62	2.69	3.48	4.92	5.05	8.02
1991	0.69	0.87	1.55	2.55	3.48	4.62	5.31	7.26
1992	0.65	0.85	1.51	2.44	3.39	4.47	5.27	6.99
1993	0.62	0.81	1.46	2.36	3.28	4.33	5.16	6.81
1994	0.59	0.78	1.40	2.28	3.17	4.18	5.02	6.65
1995	0.58	0.75	1.35	2.20	3.07	4.03	4.86	6.48
1996	0.56	0.72	1.29	2.12	2.99	3.88	4.66	6.33
1997	0.55	0.71	1.23	2.02	2.95	3.77	4.34	6.22
1998	0.53	0.72	1.16	1.88	2.88	3.98	3.71	6.24
1999	0.52	0.69	1.20	1.89	2.70	3.64	3.98	5.46
2000	0.49	0.66	1.20	1.90	2.63	3.43	3.90	5.14
2001	0.47	0.64	1.19	1.90	2.61	3.35	3.72	4.83
2002	0.45	0.60	1.19	1.90	2.59	3.43	3.55	4.45
2003	0.44	0.56	1.12	1.90	2.54	3.26	4.02	4.35
2004	0.44	0.54	1.05	1.81	2.57	3.32	4.36	5.31
2005	0.44	0.54	1.00	1.63	2.54	3.72	4.21	5.33
2006	0.44	0.59	1.03	1.70	2.65	3.53	3.88	5.20
2007	0.43	0.58	1.04	1.61	2.51	3.34	3.30	5.16
2008	0.41	0.56	1.03	1.54	2.24	3.72	4.28	5.55
2009	0.39	0.50	0.94	1.56	2.03	3.24	4.16	5.27
2010	0.39	0.48	0.96	1.57	2.16	2.83	3.50	5.21

Year	Age							
	1	2	3	4	5	6	7	8+
2011	0.39	0.46	0.91	1.60	2.08	3.01	3.54	4.87
2012	0.40	0.50	0.87	1.45	2.06	2.67	3.00	4.37
2013	0.40	0.53	0.88	1.39	1.90	2.55	2.69	4.00
2014	0.39	0.53	0.88	1.25	1.63	2.05	2.37	3.49
2015	0.40	0.51	0.88	1.15	1.50	1.88	2.19	3.64
2016	0.39	0.51	0.86	1.29	1.35	1.89	2.90	3.37
2017	0.37	0.46	0.90	1.41	1.73	2.13	2.72	3.46
2018	0.35	0.47	0.92	1.43	1.78	2.40	2.65	3.63
2019	0.41	0.47	0.90	1.41	1.92	2.33	3.03	3.33
2020	0.43	0.53	0.97	1.41	1.98	2.32	2.85	3.15
2021	0.44	0.55	0.96	1.46	1.85	2.22	2.70	2.98
2022	0.44	0.61	0.99	1.48	1.86	2.27	2.69	2.99
2023	0.39	0.59	1.01	1.45	1.81	2.07	2.62	3.21
2024	0.41	0.56	0.96	1.43	1.87	2.00	2.40	3.07

Table 20.8. Turbot in Subarea 4. Natural mortality-at-age and maturity at age.

Age	1	2	3	4	5	6	7	8+
natural mortality	0.61	0.46	0.38	0.32	0.30	0.29	0.28	0.27
maturity	0	0.01	0.58	0.98	1	1	1	1

Table 20.9. Turbot in Subarea 4. Fraction of harvest before spawning and fraction of natural mortality before spawning.

Age	1	2	3	4	5	6	7	8+
Harvest	0	0	0	0	0	0	0	0
Natural mortality	0	0	0	0	0	0	0	0

Table 20.10. Turbot in Subarea 4. BTS+COAST survey index

Year	Age						
	1	2	3	4	5	6	7
1991	12.10	64.00	5.00	0.57	0.84	0.60	1.84

Year	Age						
	1	2	3	4	5	6	7
1992	16.20	38.00	11.90	0.17	0.71	0.55	0.73
1993	18.50	32.00	3.60	1.37	0.26	0.48	0.53
1994	26.40	46.00	2.40	0.30	0.85	0.25	0.32
1995	31.90	20.00	4.20	0.09	0.52	0.67	0.51
1996	7.00	44.00	2.00	0.78	0.54	0.36	0.97
1997	4.90	46.00	10.90	1.01	0.51	0.52	0.63
1998	41.90	29.00	8.40	5.37	0.72	NA	1.30
1999	35.60	40.00	7.00	3.22	0.29	0.13	0.18
2000	65.20	28.00	22.20	8.08	0.56	0.51	0.42
2001	20.60	56.00	4.10	4.60	0.17	0.11	3.22
2002	63.20	22.00	6.00	1.64	0.35	0.46	0.19
2003	64.40	44.00	2.80	2.21	0.81	0.76	0.76
2004	65.50	26.00	15.00	1.20	4.78	2.64	1.21
2005	36.70	65.00	26.20	5.29	1.98	NA	1.10
2006	80.60	42.00	13.40	3.83	1.80	3.38	0.28
2007	48.60	56.00	29.90	12.78	2.29	2.22	0.19
2008	38.60	59.00	27.10	6.75	3.70	0.72	4.10
2009	37.50	23.00	26.60	20.50	10.00	1.37	0.76
2010	39.10	30.00	7.00	1.74	2.76	0.99	0.30
2011	75.10	45.00	5.80	0.49	4.13	0.94	1.27
2012	37.50	57.00	26.90	12.92	2.16	3.20	6.29
2013	25.10	33.00	29.90	5.07	1.43	2.17	1.00
2014	71.50	17.00	19.70	14.32	1.85	1.34	0.32
2015	138.70	62.00	13.30	2.19	5.66	0.78	0.73
2016	34.30	96.00	32.90	0.32	1.16	1.29	1.52
2017	97.50	32.00	51.40	9.34	0.87	NA	1.58
2018	59.70	52.00	16.00	18.07	7.52	0.46	0.94
2019	126.00	50.00	27.70	1.35	5.44	0.20	1.02

Year	Age						
	1	2	3	4	5	6	7
2020	73.60	59.00	25.40	4.74	0.65	1.99	0.79
2021	38.30	32.00	20.90	9.61	1.88	NA	1.00
2022	54.50	22.00	19.70	4.83	1.36	0.56	0.39
2023	60.10	48.00	22.30	19.22	3.94	1.42	0.28
2024	89.20	35.00	41.70	4.64	2.99	2.69	0.64

Table 20.11. Turbot in Subarea 4. BSAS survey index

Year	Age							
	1	2	3	4	5	6	7	8
2019	24086	14658	4952	878	2439	644	258	190
2020	9655	12609	5408	1406	465	990	77	59
2021	7419	9053	4610	1975	690	103	303	118
2022	6682	5303	3619	1030	659	160	14	473
2023	13604	9488	2514	2005	661	335	63	31
2024	11661	9892	5343	1035	916	991	275	146

Table 20.12. Turbot in Subarea 4. Fbar (Ages 2–6) estimates from the assessment model.

Year	Low	Fbar	High
1981	0.203	0.261	0.336
1982	0.210	0.269	0.345
1983	0.243	0.311	0.398
1984	0.278	0.354	0.450
1985	0.328	0.419	0.535
1986	0.310	0.398	0.512
1987	0.274	0.348	0.443
1988	0.291	0.367	0.464
1989	0.368	0.467	0.593
1990	0.504	0.677	0.908
1991	0.457	0.638	0.889

Year	Low	Fbar	High
1992	0.403	0.574	0.819
1993	0.366	0.532	0.775
1994	0.328	0.474	0.686
1995	0.294	0.424	0.611
1996	0.273	0.390	0.557
1997	0.284	0.402	0.567
1998	0.400	0.522	0.682
1999	0.348	0.478	0.658
2000	0.336	0.474	0.669
2001	0.350	0.485	0.672
2002	0.399	0.551	0.759
2003	0.533	0.664	0.826
2004	0.392	0.494	0.621
2005	0.338	0.430	0.546
2006	0.259	0.330	0.421
2007	0.253	0.320	0.403
2008	0.281	0.355	0.448
2009	0.254	0.313	0.387
2010	0.295	0.362	0.443
2011	0.263	0.321	0.390
2012	0.238	0.288	0.350
2013	0.228	0.275	0.332
2014	0.271	0.326	0.393
2015	0.310	0.372	0.445
2016	0.375	0.456	0.554
2017	0.265	0.322	0.390
2018	0.357	0.431	0.519
2019	0.380	0.459	0.555
2020	0.411	0.503	0.617

Year	Low	Fbar	High
2021	0.411	0.524	0.669
2022	0.231	0.306	0.404
2023	0.217	0.293	0.396
2024	0.180	0.256	0.363
2025			

Table 20.13. Turbot in Subarea 4. Total and Spawning Stock Biomass estimates from the assessment model (tonnes).

Year	Low	TSB	High	Low	SSB	High
1981	16179	32810	42707	17299	24013	33333
1982	15092	30776	39793	15203	21078	29223
1983	15380	31806	40800	12915	17939	24919
1984	16454	31021	39156	11683	16214	22501
1985	16018	26745	33379	11669	15745	21246
1986	13779	24372	30433	10876	14635	19692
1987	12445	23482	29270	9393	12905	17729
1988	11819	23099	28679	9010	12326	16861
1989	12088	21006	26520	8925	12086	16367
1990	11567	18735	24895	7067	9788	13556
1991	10609	13188	19013	4058	6280	9720
1992	9919	10076	14786	3145	4838	7443
1993	9013	8772	12866	2490	3780	5740
1994	8089	8346	12202	2020	3015	4502
1995	7769	9787	14220	2014	2927	4256
1996	7539	10092	14053	2178	3066	4317
1997	7600	10124	13524	2881	3883	5234
1998	7677	11113	14957	3288	4290	5599
1999	7476	10489	14764	2642	3742	5300
2000	8087	11878	16802	2803	3962	5601
2001	7821	11016	14949	2820	3867	5302

Year	Low	TSB	High	Low	SSB	High
2002	7577	12213	15347	3285	4234	5459
2003	7394	12077	14918	3034	3776	4699
2004	7399	12656	15977	2697	3465	4451
2005	7464	12914	16224	2770	3653	4818
2006	25207	15540	19732	3289	4283	5576
2007	23802	15637	19258	3959	5110	6594
2008	24795	14114	17062	5081	6455	8199
2009	24576	13416	16328	5407	6860	8704
2010	21430	14413	17637	5210	6608	8380
2011	19519	16441	20437	4901	6220	7895
2012	18838	16195	19515	5062	6274	7777
2013	18605	14864	17542	5953	7223	8765
2014	16638	15516	18353	6081	7307	8779
2015	14099	16643	19929	5031	6033	7234
2016	9147	14529	17030	5163	6142	7307
2017	6866	14149	16639	6171	7381	8827
2018	5981	14433	16971	6212	7494	9042
2019	5708	15051	18062	5130	6140	7349
2020	6736	13816	16534	4709	5671	6829
2021	7248	11788	14365	4400	5424	6687
2022	7579	11308	14290	3686	4714	6028
2023	8256	11801	15384	3547	4658	6116
2024	7451	13623	19199	3601	4950	6804
2025		13784			5817	

Table 20.14. Turbot in Subarea 4. Recruitment (Age 1) estimates from the assessment model (units: thousands).

Year	Low	Value	High
1981	5129.52	7120.78	9885.06
1982	8054.86	10911.33	14780.79

Year	Low	Value	High
1983	12223.01	16663.74	22717.84
1984	9024.85	12324.58	16830.78
1985	4690.70	6392.44	8711.54
1986	5882.46	7933.55	10699.83
1987	6463.31	8752.74	11853.13
1988	4941.57	6876.47	9568.99
1989	3718.52	5689.47	8705.11
1990	3896.91	6754.68	11708.16
1991	2387.34	4082.06	6979.84
1992	2130.42	3605.74	6102.71
1993	2805.06	4617.84	7602.16
1994	3037.79	4833.99	7692.27
1995	4839.00	7447.40	11461.83
1996	4032.35	5952.94	8788.29
1997	3287.93	4923.62	7373.05
1998	4743.80	7435.80	11655.45
1999	3930.13	6258.49	9966.24
2000	6232.75	9602.19	14793.16
2001	4319.78	6433.19	9580.55
2002	8104.37	10739.96	14232.67
2003	7729.92	10228.74	13535.33
2004	9638.55	12599.51	16470.09
2005	8507.48	11142.91	14594.74
2006	11815.42	15536.17	20428.60
2007	8321.57	11023.22	14601.97
2008	5003.79	6816.66	9286.33
2009	7271.58	9543.06	12524.10
2010	9475.18	12370.68	16151.02
2011	11937.56	16085.16	21673.80

Year	Low	Value	High
2012	8426.21	11071.17	14546.37
2013	5300.62	7095.91	9499.26
2014	10186.95	13392.97	17608.00
2015	11443.82	15539.58	21101.22
2016	4618.70	6424.40	8936.03
2017	6854.27	9038.79	11919.54
2018	8615.79	11308.73	14843.37
2019	9386.64	12703.76	17193.10
2020	6172.43	8177.40	10833.63
2021	4780.03	6508.79	8862.77
2022	5959.58	8467.04	12029.49
2023	6376.50	9514.08	14195.51
2024	6845.25	11822.41	20418.46
2025*		8529	

Table 20.15. Turbot in Subarea 4. Fishing mortality-at-age as estimated by the assessment model.

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.001	0.069	0.427	0.384	0.221	0.204	0.155	0.155
1982	0.001	0.069	0.422	0.389	0.238	0.228	0.178	0.178
1983	0.001	0.092	0.468	0.431	0.290	0.272	0.209	0.209
1984	0.002	0.134	0.540	0.473	0.330	0.293	0.209	0.209
1985	0.002	0.160	0.609	0.544	0.428	0.354	0.229	0.229
1986	0.002	0.163	0.583	0.520	0.400	0.326	0.214	0.214
1987	0.003	0.175	0.557	0.461	0.307	0.242	0.171	0.171
1988	0.004	0.219	0.600	0.457	0.314	0.248	0.183	0.183
1989	0.006	0.281	0.731	0.557	0.448	0.318	0.233	0.233
1990	0.012	0.431	0.953	0.720	0.725	0.553	0.420	0.420
1991	0.011	0.400	0.927	0.701	0.659	0.501	0.390	0.390

Year	Age							
	1	2	3	4	5	6	7	8+
1992	0.010	0.362	0.858	0.656	0.566	0.430	0.345	0.345
1993	0.009	0.338	0.817	0.622	0.505	0.380	0.309	0.309
1994	0.007	0.291	0.747	0.575	0.433	0.325	0.269	0.269
1995	0.006	0.249	0.672	0.532	0.381	0.285	0.239	0.239
1996	0.005	0.213	0.613	0.501	0.354	0.267	0.226	0.226
1997	0.005	0.208	0.602	0.514	0.392	0.292	0.245	0.245
1998	0.008	0.269	0.703	0.625	0.590	0.425	0.342	0.342
1999	0.009	0.275	0.653	0.579	0.522	0.364	0.302	0.302
2000	0.012	0.323	0.658	0.565	0.494	0.330	0.280	0.280
2001	0.017	0.382	0.681	0.570	0.480	0.312	0.270	0.270
2002	0.028	0.508	0.747	0.618	0.544	0.336	0.296	0.296
2003	0.046	0.671	0.843	0.696	0.689	0.418	0.367	0.367
2004	0.044	0.613	0.711	0.547	0.377	0.220	0.182	0.182
2005	0.042	0.559	0.633	0.472	0.289	0.195	0.179	0.179
2006	0.030	0.439	0.491	0.361	0.198	0.160	0.173	0.173
2007	0.022	0.395	0.445	0.345	0.220	0.193	0.185	0.185
2008	0.023	0.388	0.445	0.370	0.313	0.257	0.205	0.205
2009	0.023	0.400	0.429	0.333	0.223	0.181	0.160	0.160
2010	0.026	0.426	0.456	0.369	0.289	0.269	0.233	0.233
2011	0.021	0.373	0.414	0.341	0.245	0.230	0.212	0.212
2012	0.016	0.305	0.380	0.326	0.216	0.214	0.208	0.208
2013	0.014	0.271	0.361	0.326	0.211	0.206	0.195	0.195
2014	0.008	0.200	0.355	0.373	0.331	0.372	0.360	0.360
2015	0.007	0.194	0.376	0.417	0.432	0.439	0.412	0.412
2016	0.007	0.188	0.413	0.500	0.636	0.543	0.493	0.493
2017	0.004	0.127	0.334	0.401	0.398	0.349	0.350	0.350
2018	0.011	0.206	0.428	0.494	0.545	0.481	0.448	0.448
2019	0.019	0.249	0.461	0.515	0.569	0.503	0.440	0.440

Year	Age							
	1	2	3	4	5	6	7	8+
2020	0.032	0.295	0.488	0.540	0.651	0.541	0.463	0.463
2021	0.032	0.264	0.470	0.528	0.704	0.656	0.554	0.554
2022	0.018	0.165	0.333	0.354	0.342	0.334	0.336	0.336
2023	0.018	0.157	0.320	0.338	0.320	0.332	0.351	0.351
2024	0.019	0.147	0.302	0.311	0.258	0.261	0.289	0.289

Table 20.16. Turbot in Subarea 4. Population abundance (stock numbers at age) as estimated by the assessment model (units: thousands).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	7120.78	5648.69	2303.66	1884.51	2705.92	1095.74	556.22	915.72
1982	10911.33	3777.62	3331.35	1005.96	924.86	1612.69	673.86	971.42
1983	16663.74	5935.25	2254.46	1478.46	487.02	539.11	966.29	1051.61
1984	12324.58	9449.96	3454.55	1001.67	679.89	265.43	309.85	1224.39
1985	6392.44	6835.02	5104.65	1410.94	477.19	361.35	147.39	937.29
1986	7933.55	3298.93	3830.58	1818.49	584.70	236.16	187.91	658.40
1987	8752.74	4336.78	1722.69	1539.62	741.24	283.65	128.44	518.04
1988	6876.47	4875.94	2261.65	689.84	696.36	393.87	166.91	417.03
1989	5689.47	3685.58	2423.20	861.87	349.40	379.10	224.18	368.92
1990	6754.68	3013.71	1782.72	777.65	356.87	170.05	215.48	357.24
1991	4082.06	3802.11	1177.83	469.83	274.69	128.01	73.46	285.62
1992	3605.74	2191.35	1649.41	306.21	171.35	105.39	58.19	183.88
1993	4617.84	1910.71	944.30	487.96	114.37	72.56	51.43	129.95
1994	4833.99	2541.47	830.58	283.46	193.15	51.33	37.29	101.17
1995	7447.40	2547.92	1200.38	264.39	117.02	93.92	27.94	81.02
1996	5952.94	4200.15	1221.24	423.37	115.39	59.55	53.41	65.80
1997	4923.62	3382.11	2202.12	454.65	189.03	61.23	34.38	72.84
1998	7435.80	2723.41	1788.17	855.62	198.80	97.07	34.80	64.29
1999	6258.49	4132.38	1370.70	597.10	326.25	81.34	47.62	53.46

Year	Age							
	1	2	3	4	5	6	7	8+
2000	9602.19	3241.67	2030.98	513.93	234.89	145.05	42.65	57.15
2001	6433.19	5351.71	1422.38	734.33	215.11	104.19	79.39	57.95
2002	10739.96	3313.60	2361.71	483.43	300.31	103.48	57.25	79.56
2003	10228.74	5825.81	1208.68	801.18	183.56	131.93	58.06	78.33
2004	12599.51	5212.50	1908.78	351.77	298.64	64.70	66.04	70.89
2005	11142.91	6640.48	1736.91	649.45	142.30	151.16	36.99	92.34
2006	15536.17	5728.57	2440.40	588.65	280.95	76.64	92.60	83.54
2007	11023.22	8347.31	2295.74	1048.22	289.91	171.70	48.71	107.90
2008	6816.66	6025.79	3455.95	1018.39	532.57	175.93	103.79	95.19
2009	9543.06	3409.70	2732.30	1567.02	535.03	256.01	98.18	121.25
2010	12370.68	5166.88	1325.65	1242.01	809.81	328.91	156.74	135.25
2011	16085.16	6621.09	2184.66	537.81	653.80	449.37	184.70	168.58
2012	11071.17	8923.01	2807.76	1043.31	277.68	383.31	274.44	206.74
2013	7095.91	6017.01	4330.21	1271.15	554.02	168.58	234.77	281.03
2014	13392.97	3552.44	2795.15	2204.91	639.95	347.45	106.07	327.85
2015	15539.58	7409.76	1845.66	1324.06	1120.74	340.40	177.44	234.62
2016	6424.40	8833.12	3793.33	874.67	653.95	548.74	163.44	209.94
2017	9038.79	3223.58	4923.95	1683.89	387.75	245.86	245.21	174.94
2018	11308.73	4937.95	1726.44	2500.05	792.77	197.10	132.14	226.39
2019	12703.76	6024.66	2551.00	746.64	1099.90	331.32	92.86	175.74
2020	8177.40	6531.17	2925.04	1068.21	317.44	460.40	146.72	128.79
2021	6508.79	4154.98	2862.65	1247.77	432.24	119.76	205.16	134.80
2022	8467.04	3316.28	2013.61	1142.47	516.04	151.69	44.37	151.33
2023	9514.08	4648.50	1757.22	1006.47	584.29	270.69	83.45	104.01
2024	11822.41	4942.36	2563.39	875.35	515.34	318.86	143.91	105.95

Table 20.17. Turbot in Subarea 4. Predicted catch numbers-at-age (units: thousands).

Year	Age							
	1	2	3	4	5	6	7	8+
1981	2.99	304.02	675.02	518.89	465.29	176.60	70.03	115.48
1982	4.57	202.90	967.48	279.69	170.12	287.38	96.35	139.12
1983	11.07	417.26	711.95	447.29	106.71	112.57	160.09	174.51
1984	15.27	950.78	1221.01	326.77	166.21	59.05	51.29	203.01
1985	10.63	811.98	1975.37	513.47	144.88	94.42	26.54	169.08
1986	13.59	398.79	1435.38	638.80	168.16	57.53	31.86	111.82
1987	16.92	560.95	623.32	492.21	170.28	53.43	17.75	71.71
1988	20.49	772.51	865.27	218.72	163.32	75.74	24.56	61.46
1989	25.10	729.45	1070.71	319.10	110.17	90.53	40.97	67.52
1990	59.18	858.85	939.15	347.77	161.75	63.68	65.15	108.18
1991	33.29	1019.34	609.65	206.07	116.36	44.39	20.91	81.44
1992	26.14	539.82	812.60	128.12	64.83	32.37	14.94	47.29
1993	31.32	443.90	450.25	196.37	39.64	20.12	12.04	30.47
1994	26.95	519.49	372.33	107.61	59.21	12.48	7.73	21.01
1995	33.55	453.27	499.62	94.51	32.29	20.41	5.22	15.17
1996	21.81	650.12	475.02	144.49	29.99	12.21	9.48	11.70
1997	18.25	512.73	845.21	158.32	53.39	13.59	6.58	13.96
1998	44.01	518.63	768.39	345.35	77.57	29.55	8.88	16.43
1999	40.46	802.66	558.47	227.74	115.97	21.80	10.93	12.29
2000	85.10	725.48	832.44	192.53	80.01	35.77	9.15	12.28
2001	79.25	1381.21	597.59	276.77	71.60	24.48	16.55	12.10
2002	220.20	1077.15	1058.82	193.68	110.21	25.89	12.90	17.96
2003	347.58	2338.21	588.46	349.81	80.28	39.62	15.71	21.23
2004	410.23	1956.37	827.19	128.52	81.74	11.16	9.65	10.38
2005	342.32	2323.24	691.62	211.58	31.08	23.41	5.31	13.27
2006	343.39	1657.63	801.03	153.74	43.86	9.92	12.90	11.66
2007	181.33	2214.36	696.68	263.95	49.66	26.28	7.22	16.01
2008	115.10	1573.91	1048.77	271.50	124.59	34.92	16.90	15.53

Year	Age							
	1	2	3	4	5	6	7	8+
2009	162.48	914.24	804.29	382.95	92.73	36.97	12.77	15.80
2010	242.48	1459.22	409.81	330.49	176.92	67.84	28.61	24.73
2011	255.05	1672.55	624.82	134.07	123.48	80.80	30.98	28.32
2012	133.74	1900.33	748.16	250.26	46.84	64.49	45.21	34.11
2013	74.83	1156.43	1105.57	304.47	91.57	27.44	36.51	43.78
2014	76.43	519.09	704.02	591.90	157.17	94.64	28.26	87.50
2015	77.32	1053.78	487.61	390.08	343.31	106.31	52.87	70.02
2016	31.86	1221.97	1082.65	297.94	269.85	202.64	56.20	72.30
2017	24.31	308.39	1177.06	480.16	111.08	63.46	63.86	45.63
2018	95.96	742.02	506.93	843.37	291.49	66.25	42.13	72.29
2019	180.93	1072.65	795.68	260.26	418.10	115.29	29.17	55.29
2020	191.40	1350.00	955.43	386.63	133.28	169.45	48.06	42.25
2021	154.29	778.58	907.38	443.53	191.82	50.88	77.19	50.80
2022	112.77	407.22	480.00	293.77	130.05	37.75	11.15	38.10
2023	127.00	543.07	404.75	248.65	139.12	67.04	21.76	27.17
2024	163.77	545.25	561.23	201.38	101.80	64.18	31.79	23.44

Table 20.18. Turbot in Subarea 4. Catch-at-age residuals.

Year	Age							
	1	2	3	4	5	6	7	8+
1981	0.000	-0.055	0.458	1.440	2.035	0.357	2.862	0.000
1982	0.000	0.010	0.507	-1.262	-1.341	-2.094	-0.241	0.185
1983	0.000	1.447	-0.042	0.453	0.144	-0.151	0.085	0.025
1984	0.000	2.193	-0.198	0.249	-0.434	-0.276	-0.177	-0.914
1985	0.000	-0.517	-0.048	0.970	1.295	-0.106	-0.762	-0.830
1986	0.000	-1.316	-0.347	-0.808	0.062	0.369	-0.570	-0.068
1987	0.303	0.716	-1.845	0.781	-1.456	-0.939	0.144	-0.497
1988	0.783	0.782	-1.091	-1.338	0.268	0.028	0.215	0.049

Year	Age							
	1	2	3	4	5	6	7	8+
1989	0.000	-0.664	0.169	0.589	2.544	-0.553	-0.658	-0.306
1990	0.485	0.355	0.134	-1.029	0.425	1.603	2.068	-0.286
1991–1997	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1998	0.000	1.511	2.068	2.240	1.339	1.086	0.389	-0.285
1999–2002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2003	1.478	0.989	-1.157	1.203	-0.135	0.953	2.056	0.390
2004	0.660	-0.456	-0.777	-0.019	-0.286	-1.682	-1.218	-1.706
2005	0.248	-0.283	-0.626	0.221	-1.063	0.442	-1.747	2.205
2006	1.047	-0.627	-0.716	-2.215	-1.017	-0.170	1.105	1.326
2007	-1.623	0.956	-0.809	1.048	0.169	1.297	-0.190	-0.715
2008	-0.023	-0.481	-0.793	0.183	2.806	1.000	-1.781	-1.660
2009	-0.388	-0.457	1.305	0.505	-0.433	-2.715	-0.268	0.570
2010	0.817	0.453	-1.215	0.340	0.505	2.277	0.118	-0.966
2011	-0.106	0.755	0.384	-1.214	0.478	-0.253	0.094	-1.015
2012	0.000	-0.141	0.343	1.331	-0.809	0.337	1.614	-1.736
2013	-0.148	-0.038	0.699	0.155	-0.209	-0.085	0.288	-2.115
2014	-0.691	-2.168	0.503	2.327	0.447	2.385	1.458	0.695
2015	-0.671	1.364	0.857	-0.105	0.657	-0.148	-0.710	0.714
2016	0.000	-0.402	0.718	1.455	1.922	-0.642	-0.497	0.332
2017	-2.115	-1.816	1.347	-0.433	-0.552	-1.376	1.096	0.279
2018	2.599	0.106	-0.553	0.756	-0.697	0.945	0.416	0.230
2019	0.953	-0.138	0.186	-0.632	-0.542	-0.130	-0.676	0.752
2020	0.493	-0.105	-0.908	-0.329	0.570	-0.560	-0.481	-0.037
2021	0.000	-1.214	-0.596	0.384	0.076	0.886	-0.171	0.914
2022	0.722	-1.434	-1.075	-2.525	-0.855	-1.601	-0.403	0.249
2023	0.059	0.131	-0.217	-0.222	-0.035	0.460	0.788	-0.325
2024	0.988	-1.130	0.157	-0.147	-0.847	-0.478	-0.989	1.722

Table 20.19. Turbot in Subarea 4. Predicted index-at-age BTS+COAST.

Year	Age						
	1	2	3	4	5	6	7
1991	19.582	31.513	5.642	1.405	0.907	0.451	1.381
1992	17.312	18.663	8.290	0.945	0.604	0.391	0.961
1993	22.182	16.552	4.886	1.543	0.421	0.279	0.738
1994	23.246	22.751	4.517	0.926	0.748	0.205	0.580
1995	35.850	23.501	6.879	0.891	0.470	0.385	0.466
1996	28.679	39.728	7.297	1.458	0.472	0.248	0.515
1997	23.719	32.100	13.259	1.551	0.754	0.250	0.457
1998	35.747	24.770	10.028	2.700	0.690	0.361	0.394
1999	30.071	37.424	7.963	1.946	1.187	0.316	0.413
2000	46.032	28.369	11.755	1.691	0.872	0.577	0.415
2001	30.739	44.920	8.100	2.409	0.806	0.420	0.574
2002	50.916	25.454	12.842	1.532	1.076	0.410	0.562
2003	47.861	39.892	6.140	2.404	0.593	0.493	0.533
2004	59.036	37.192	10.641	1.173	1.203	0.278	0.609
2005	52.305	49.229	10.235	2.282	0.610	0.661	0.577
2006	73.540	46.192	15.888	2.238	1.284	0.343	0.789
2007	52.463	69.433	15.439	4.028	1.305	0.752	0.696
2008	32.429	50.382	23.241	3.847	2.245	0.736	0.871
2009	45.393	28.260	18.589	6.073	2.403	1.131	0.992
2010	58.698	42.043	8.850	4.695	3.471	1.366	1.254
2011	76.598	55.955	15.019	2.073	2.891	1.917	1.539
2012	52.912	79.089	19.771	4.064	1.253	1.655	2.103
2013	33.963	54.618	30.900	4.953	2.509	0.732	2.275
2014	64.398	33.913	20.028	8.312	2.662	1.342	1.704
2015	74.771	71.030	13.032	4.837	4.343	1.254	1.560
2016	30.913	85.019	26.098	3.014	2.195	1.878	1.335
2017	43.586	32.405	35.813	6.223	1.539	0.965	1.660
2018	54.233	46.933	11.756	8.653	2.837	0.705	1.323

Year	Age						
	1	2	3	4	5	6	7
2019	60.589	55.560	16.970	2.546	3.870	1.166	0.997
2020	38.659	58.311	19.084	3.577	1.054	1.578	1.005
2021	30.761	37.921	18.917	4.216	1.383	0.379	1.164
2022	40.418	32.438	14.657	4.364	2.132	0.602	0.782
2023	45.415	45.747	12.909	3.889	2.451	1.075	0.741
2024	56.407	48.960	19.076	3.447	2.258	1.331	1.031

Table 20.20. Turbot in Subarea 4. Index-at-age residuals BTS+COAST

Year	Age						
	1	2	3	4	5	6	7
1991	-1.684	1.107	-1.327	-1.540	0.065	0.266	-0.051
1992	-0.271	0.995	0.445	-2.710	0.965	0.392	-0.617
1993	0.001	0.803	-0.569	0.260	-0.372	0.910	-0.550
1994	0.452	1.109	-1.111	-1.095	1.025	0.463	-0.624
1995	0.448	-0.186	-0.231	-2.449	1.484	1.027	0.178
1996	-1.705	1.393	-1.201	0.379	1.157	0.668	0.712
1997	-1.291	2.108	0.679	0.038	0.063	1.150	0.128
1998	1.425	0.554	0.121	1.245	0.029	0.000	1.459
1999	0.479	0.481	0.316	0.976	-1.894	-0.377	-0.591
2000	1.008	-0.463	1.366	2.427	-0.744	0.425	0.307
2001	-1.003	0.521	-1.368	1.214	-1.906	-0.533	2.558
2002	0.999	-0.761	-0.776	0.230	-0.955	1.177	-0.988
2003	0.201	-0.007	-0.945	0.557	0.628	0.859	0.773
2004	0.048	-0.703	0.902	0.324	2.539	2.290	0.633
2005	-0.799	0.553	1.166	1.034	1.430	0.000	0.365
2006	0.495	0.022	0.174	0.874	0.180	2.125	-2.125
2007	-0.168	-0.183	1.224	1.445	0.140	0.573	-2.181
2008	-0.155	0.624	0.279	0.659	-0.168	-0.551	1.336

Year	Age						
	1	2	3	4	5	6	7
2009	-0.243	-0.799	0.579	1.540	1.571	-0.698	-1.043
2010	-0.173	-0.243	-0.615	-1.493	-0.430	-0.719	-2.127
2011	0.734	-0.046	-0.923	-1.792	1.027	-0.929	-0.511
2012	-0.537	0.375	0.769	1.820	0.601	0.526	0.777
2013	-1.050	-0.271	0.528	-0.070	-0.774	1.267	-1.393
2014	1.592	-0.761	0.062	0.677	-0.995	-0.136	-1.961
2015	1.736	-0.183	0.109	-1.206	0.374	-0.667	-0.717
2016	-0.835	0.754	0.239	-3.347	-0.327	0.135	0.487
2017	1.766	-0.584	0.579	0.438	-0.684	0.000	0.211
2018	-0.334	-0.039	0.196	0.942	1.104	-0.709	-0.143
2019	0.573	-0.734	0.420	-1.242	0.506	-2.150	0.556
2020	-0.477	-0.639	0.219	-0.008	-1.170	0.164	-0.653
2021	-0.328	-0.561	-0.279	0.948	-0.196	0.000	-0.266
2022	0.400	-0.725	0.528	-0.019	-0.864	-0.206	-1.005
2023	0.834	0.204	0.902	2.364	0.437	0.194	-1.234

Table 20.21. Turbot in Subarea 4. Predicted index-at-age BSAS

Year	Age							
	1	2	3	4	5	6	7	8
2019	15111.096	11798.074	4301.780	921.493	1572.941	491.547	85.335	161.911
2020	9641.542	12382.168	4837.832	1294.723	428.528	665.107	132.610	116.701
2021	7671.957	8052.374	4795.519	1525.784	562.360	159.553	174.004	114.622
2022	10080.427	6888.061	3715.437	1579.326	866.481	253.568	43.865	149.999
2023	11326.650	9714.158	3272.446	1407.456	996.426	453.082	81.641	102.015
2024	14067.974	10396.577	4835.669	1247.602	917.934	560.972	147.075	108.551

Table 20.22. Turbot in Subarea 4. Index-at-age residuals BSAS

Year	Age							
	1	2	3	4	5	6	7	8
2019	0.016	-0.142	-0.109	-0.257	0.791	0.415	2.110	-0.162

Age								
Year	1	2	3	4	5	6	7	8
2020	-1.777	-0.574	0.025	-0.194	-0.253	0.420	-1.512	-1.519
2021	-0.819	-0.126	-0.704	0.125	-0.220	-1.435	1.087	-0.072
2022	-1.168	-0.463	0.065	-0.842	-0.573	-0.993	-2.369	2.327
2023	0.451	0.097	-0.481	0.750	-0.731	-0.529	-0.335	-2.176
2024	-0.864	-0.086	0.259	-0.291	0.195	1.436	1.477	0.654

Table 20.23. Turbot in Subarea 4. Fit parameters.

Name	value	std.dev
logFpar	-4.903	0.105
logFpar	-4.184	0.129
logFpar	-4.422	0.131
logFpar	-5.09	0.137
logFpar	-5.036	0.147
logFpar	-5.093	0.18
logFpar	0.616	0.178
logFpar	1.174	0.226
logFpar	1.114	0.226
logFpar	0.801	0.23
logFpar	0.972	0.239
logFpar	0.95	0.259
logFpar	0.42	0.28
logSdLogFsta	-0.517	0.339
logSdLogFsta	-1.186	0.187
logSdLogFsta	-1.735	0.2
logSdLogFsta	-1.12	0.168
logSdLogN	-0.899	0.17
logSdLogN	-2.321	0.396
logSdLogObs	-0.354	0.24

Name	value	std.dev
logSdLogObs	-1.577	0.14
logSdLogObs	-1.159	0.145
logSdLogObs	-0.637	0.145
logSdLogObs	-0.368	0.069
logSdLogObs	-0.129	0.095
logSdLogObs	-1.151	0.367
logSdLogObs	-0.712	0.123
transfIRARdist	1.058	0.303

Table 20.24. Turbot in Subarea 4. Negative Log-Likelihood.

364.8018

Table 20.25. Comparison between the WGNSSK 2025 and 2024 estimates for recruitment, catch, fishing pressure (F) and targeted fishing pressure for total allowable catch (Target F for TAC). Note that values are not comparable due to the 2025 benchmark.

Parameter	Year*	Current Assessment (2025)	Previous assessment (2024)
Assumed recruitment	2024	11822 thousand	4370 thousand
	2025	8259 thousand	4372 thousand
Catch	2024	1831 t	1843 t
F	2024	0.26 (estimated F)	0.302 (assumed F)
Target F for TAC (F_{MSY})	2025	0.386	0.361

*Note '2024' = Intermediate year in the previous assessment; '2025' = advice year in the previous assessment